

Azure Active Directory Domain Controller

Terraform Deployment - Viewer Guide

Author	Dalla Samuel (CyberJKD)
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Course	Cloud System Admin Accelerator - Azure Administrator (AZ-104) Labs, Cloud Tech Techniques
Video	I Deployed a Full Active Directory Domain Controller with One Terraform Command Azure Lab
Roadmap	Phase 06 - Cloud Tech Techniques - Cloud System Admin Accelerator

How to Use This Guide

This document mirrors the video walkthrough step for step.

Each ■ marker is a checkpoint - a point in the video where a real decision, command, or fix happened. Pause the video at each one, run the same command on your own machine, and compare your output (and the screenshot below each checkpoint) against the "Expect" line before moving on.

This lab was not a clean, scripted run. Several real deployment failures happen along the way - SKU availability, quota limits, a region switch, and a state-drift issue. They're left in deliberately, with real screenshots of each, because working through them is most of what the lab actually teaches.

Important: this guide describes the configuration files but does not reproduce their full contents inline.

Use the actual main.tf, variables.tf, outputs.tf, and terraform.tfvars.example files included in the github repo alongside this guide (in the terraform/ folder) copy them directly rather than retyping from screenshots or asking an AI to reconstruct them.

The included files are the exact, tested version, already corrected for the multi-line string issue covered in Checkpoint 07.

Part 1 - Prerequisites

■ 00 Confirm the terminal is local

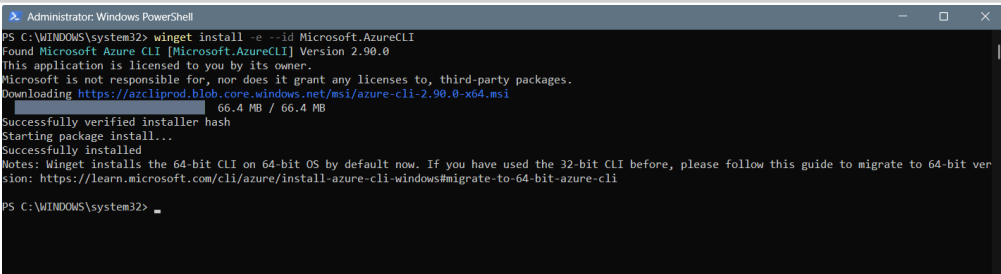
Everything in this lab runs from a PowerShell window on your own Windows machine, not inside a browser, not inside Azure Cloud Shell.

Expect: You're at a plain PS C:\WINDOWS\system32> prompt.

■ 01 Install Azure CLI

Run: `winget install -e --id Microsoft.AzureCLI` Then fully close and reopen PowerShell as Administrator before continuing - a window open during install won't see the new PATH entry.

Expect: `az --version` prints a version table (e.g. `azure-cli 2.90.0`).



```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> winget install -e --id Microsoft.AzureCLI
Found Microsoft Azure CLI [Microsoft.AzureCLI] Version 2.90.0
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://azcliproduct.blob.core.windows.net/msi/azure-cli-2.90.0-x64.msi
66.4 MB / 66.4 MB
Successfully verified installer hash
Starting package install...
Successfully installed
Notes: Winget installs the 64-bit CLI on 64-bit OS by default now. If you have used the 32-bit CLI before, please follow this guide to migrate to 64-bit version: https://learn.microsoft.com/cli/azure/install-azure-cli-windows#migrate-to-64-bit-azure-cli
PS C:\WINDOWS\system32>
```

winget installing Azure CLI

```

Administrator: Windows PowerShell
Windows PowerShell
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PS C:\WINDOWS\system32> az --version
azure-cli                        2.90.0
core                            2.90.0
telemetry                       1.1.0

Dependencies:
msal                            1.36.0
azure-mgmt-resource             24.0.0

Python location 'C:\Program Files\Microsoft SDKs\Azure\CLI2\python.exe'
Config directory 'C:\Users\dalla\.azure'
Extensions directory 'C:\Users\dalla\.azure\cliextensions'

Python (Windows) 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]

Legal docs and information: aka.ms/AzureCliLegal

Your CLI is up-to-date.
PS C:\WINDOWS\system32>

```

Fresh window confirms azure-cli 2.90.0

■ 02 Install Terraform

Run: `winget install -e --id Hashicorp.Terraform` Same rule, close and reopen the terminal after install.

Expect: `terraform -version` prints Terraform v1.16.2 on windows_amd64.

```

PS C:\WINDOWS\system32> winget install -e --id Hashicorp.Terraform
Found HashiCorp Terraform [Hashicorp.Terraform] Version 1.16.2
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://releases.hashicorp.com/terraform/1.16.2/terraform_1.16.2_windows_amd64.zip
34.9 MB / 34.9 MB
Successfully verified installer hash
Extracting archive...
Successfully extracted archive
Starting package install...
Command line alias added: "terraform"
Path environment variable modified; restart your shell to use the new value.
Successfully installed
PS C:\WINDOWS\system32>

```

winget installing Terraform

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

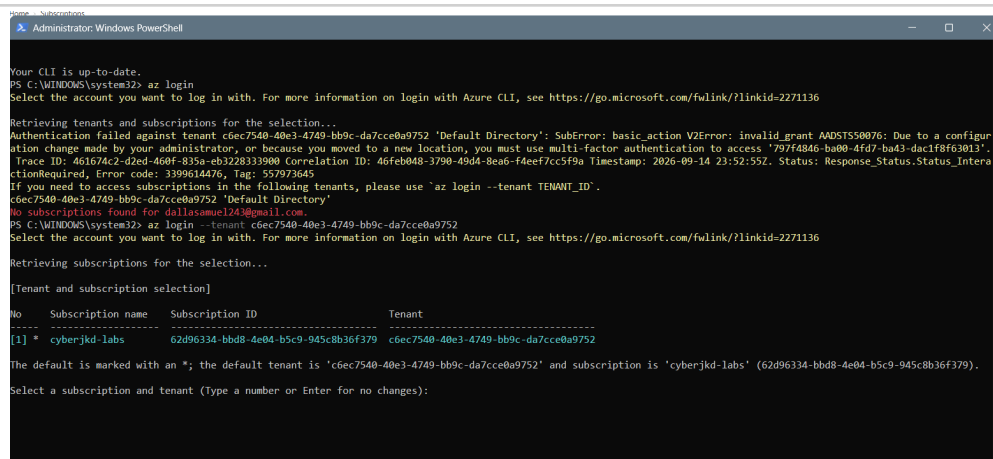
PS C:\WINDOWS\system32> terraform --version
Terraform v1.16.2
on windows_amd64
PS C:\WINDOWS\system32> _
```

Terraform v1.16.2 confirmed

■ 03 Sign in to Azure

Run `az login`. If you hit AADSTS50076 (MFA required), re-run with the tenant flag: `az login --tenant <TENANT_ID>`, and make sure you fully complete the MFA prompt before closing the browser tab.

Expect: `az account show` prints your subscription with `"isDefault": true`.



```
Administrator: Windows PowerShell

Your CLI is up-to-date.
PS C:\WINDOWS\system32> az login
Select the account you want to log in with. For more information on login with Azure CLI, see https://go.microsoft.com/fwlink/?linkid=2271136

Retrieving tenants and subscriptions for the selection...
Authentication failed against tenant c6ec7540-40e3-4749-bb9c-da7cce0a9752 'Default Directory': SubError: basic_action V2Error: invalid_grant AADSTS50076: Due to a configuration change made by your administrator, or because you moved to a new location, you must use multi-factor authentication to access '797f4846-ba00-4fd7-ba43-dac1f8f63013'.
Trace ID: 461674c2-d2ed-460f-835a-e33228339900 Correlation ID: 46feb048-3790-49d4-8ea6-f4eef7cc5f9a Timestamp: 2026-09-14 23:52:55Z. Status: Response_Status.Status_InteractionRequired, Error code: 3399614476, Tag: 557973845
If you need to access subscriptions in the following tenants, please use "az login --tenant TENANT_ID".
c6ec7540-40e3-4749-bb9c-da7cce0a9752 'Default Directory'
No subscriptions found for dallasamuel243@gmail.com.
PS C:\WINDOWS\system32> az login --tenant c6ec7540-40e3-4749-bb9c-da7cce0a9752
Select the account you want to log in with. For more information on login with Azure CLI, see https://go.microsoft.com/fwlink/?linkid=2271136

Retrieving subscriptions for the selection...

[Tenant and subscription selection]

No Subscription name Subscription ID Tenant
-----
[1] * cyberjkd-labs 62d96334-bbd8-4e04-b5c9-945c8b36f379 c6ec7540-40e3-4749-bb9c-da7cce0a9752

The default is marked with an *; the default tenant is 'c6ec7540-40e3-4749-bb9c-da7cce0a9752' and subscription is 'cyberjkd-labs' (62d96334-bbd8-4e04-b5c9-945c8b36f379).
Select a subscription and tenant (Type a number or Enter for no changes):
```

Tenant and subscription selection after fixing the MFA error

```
PS C:\WINDOWS\system32> az account show
{
  "environmentName": "AzureCloud",
  "homeTenantId": "c6ec7540-40e3-4749-bb9c-da7cce0a9752",
  "id": "62d96334-bbd8-4e04-b5c9-945c8b36f379",
  "isDefault": true,
  "managedByTenants": [],
  "name": "cyberjkd-labs",
  "state": "Enabled",
  "tenantId": "c6ec7540-40e3-4749-bb9c-da7cce0a9752",
  "user": {
    "name": "dallasamuel1243@gmail.com",
    "type": "user"
  }
}
PS C:\WINDOWS\system32> _
```

az account show confirms the active subscription

Part 2 - Scaffold the Project

■ 04 Create the project folder and files

New-Item -ItemType Directory -Force -Path "\$HOME\repos\az-ad-vm" | Out-Null
then cd "\$HOME\repos\az-ad-vm" then New-Item -ItemType File main.tf,
variables.tf, outputs.tf, terraform.tfvars

Expect: dir shows all four files sitting at 0 bytes in the new folder.

```
Windows PowerShell
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PS C:\WINDOWS\system32> terraform --version
Terraform v1.16.2
on windows_amd64
PS C:\WINDOWS\system32> New-Item -ItemType Directory -Force -Path "$HOME\repos\az-ad-vm" | Out-Null
PS C:\WINDOWS\system32> cd "$HOME\repos\az-ad-vm"
PS C:\Users\dalla\repos\az-ad-vm> New-Item -ItemType File main.tf, variables.tf, outputs.tf, terraform.tfvars

Directory: C:\Users\dalla\repos\az-ad-vm

Mode                LastWriteTime         Length Name
----                -
-a----          9/15/2026   1:09 AM             0 main.tf
-a----          9/15/2026   1:09 AM             0 variables.tf
-a----          9/15/2026   1:09 AM             0 outputs.tf
-a----          9/15/2026   1:09 AM             0 terraform.tfvars

PS C:\Users\dalla\repos\az-ad-vm> _
```

Folder created, four empty Terraform files confirmed

Part 3 - Fill In the Configuration

■ 05 Paste main.tf, variables.tf, outputs.tf

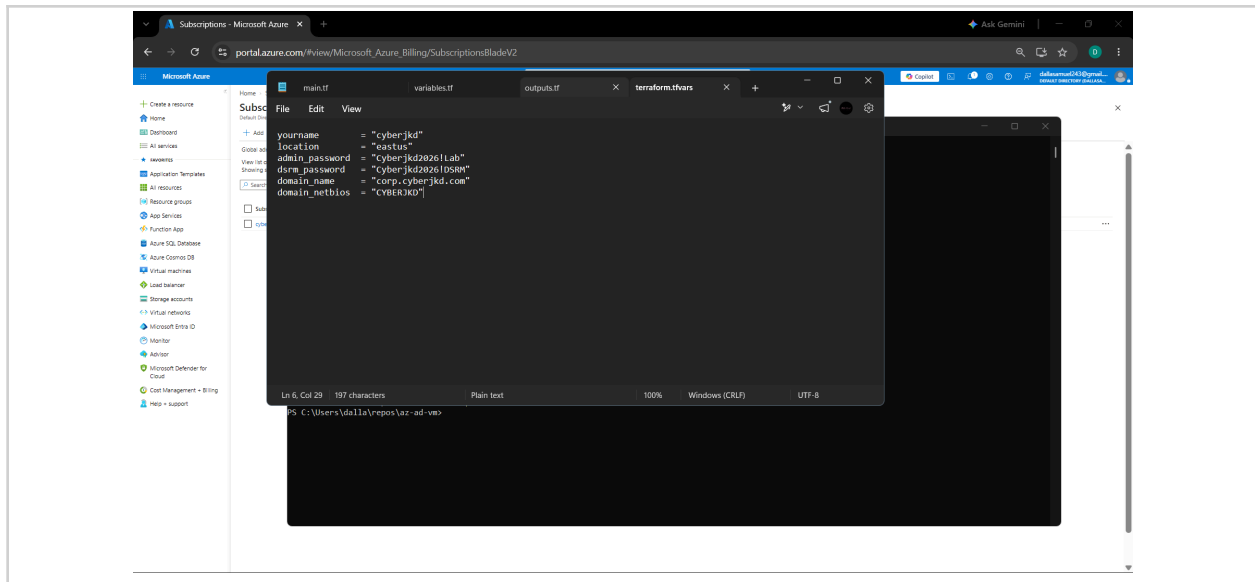
Open each with notepad <filename>, paste the matching block from the lab document, save, close. These three don't need any edits.

***Expect:** All three tabs close without a save prompt after Ctrl+S.*

■ 06 Customize terraform.tfvars

This file needs real values: your own name in place of any placeholder, your chosen domain and NetBIOS name, and two distinct passwords (admin + DSRM) meeting Azure's complexity rule 12+ characters, at least 3 of upper/lower/number/symbol.

***Expect:** The file shows your actual values, not the lab document's placeholder name.*



terraform.tfvars filled in with real project values

■ 07 Watch for the multi-line string trap

If you copy the AutoLogon content block or the Custom Script Extension command straight from a PDF, the line breaks often come with it. Terraform requires quoted strings to stay on one logical line. If terraform init throws "Invalid multi-line string," this is why - rewrite the offending block as a single line.

Expect: terraform init reports "Terraform has been successfully initialized!" with no errors

```
PS C:\Users\dalla\repos\az-ad-vm> notepad main.tf
PS C:\Users\dalla\repos\az-ad-vm> terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 3.0"...
- Installing hashicorp/azurerm v3.117.1...
- Installed hashicorp/azurerm v3.117.1 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\dalla\repos\az-ad-vm>
```

Clean init after fixing the multi-line string blocks

Part 4 - Deploy

■ 08 Preview with terraform plan

This shows every resource Terraform intends to create - nothing happens to Azure yet.

Expect: Plan summary ends with something like "Plan: 9 to add, 0 to change, 0 to destroy."

```
+ patch_assessment_mode      = "ImageDefault"
+ patch_mode                 = "AutomaticByOS"
+ platform_fault_domain      = -1
+ priority                   = "Regular"
+ private_ip_address         = (known after apply)
+ private_ip_addresses       = (known after apply)
+ provision_vm_agent         = true
+ public_ip_address          = (known after apply)
+ public_ip_addresses        = (known after apply)
+ resource_group_name        = "rg-ad-cyberjkd"
+ size                      = "Standard_D2s_v3"
+ tags                      = {
+   "project" = "ad-lab"
+ }
+ virtual_machine_id         = (known after apply)
+ vm_agent_platform_updates_enabled = false

+ additional_unattend_content {
+   content = (sensitive value)
+   setting = "Autologon"
+ }

+ os_disk {
+   caching              = "ReadWrite"
+   disk_size_gb         = 127
+   name                 = (known after apply)
+   storage_account_type = "Premium_LRS"
+   write_accelerator_enabled = false
+ }

+ source_image_reference {
+   offer      = "WindowsServer"
+   publisher  = "MicrosoftWindowsServer"
+   sku        = "2022-Datacenter"
+   version    = "latest"
+ }

+ termination_notification (known after apply)
}

Plan: 9 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ admin_username = "adadmin"
+ domain_name    = "corp.cyberjkd.com"
+ public_ip      = (known after apply)

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
PS C:\Users\dalla\repos\az-ad-vm>
```

Clean plan - 9 resources, nothing to Azure yet

■ 09 Apply and expect the first failure

terraform apply, type yes. On a first attempt it's common to hit SkuNotAvailable for the default VM size in eastus - this is a live regional capacity issue, not a mistake in your config.

Expect: An error naming a specific VM size and "is currently not available in location."

```

+ version = "latest"
}
+ termination_notification (known after apply)
}

Plan: 9 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ admin_username = "adadmin"
+ domain_name    = "corp.cyberjkd.com"
+ public_ip      = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

azure_rm_resource_group.main: Creating...
azure_rm_resource_group.main: Still creating... [00m00s elapsed]
azure_rm_resource_group.main: Creation complete after 17s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd]
azure_rm_virtual_network.main: Creating...
azure_rm_network_security_group.main: Creating...
azure_rm_public_ip.main: Creating...
azure_rm_network_security_group.main: Creation complete after 10s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkSecurityGroups/nsg-ad-cyberjkd]
azure_rm_public_ip.main: Creation complete after 10s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/publicIPAddresses/pip-ad-cyberjkd]
azure_rm_virtual_network.main: Creation complete after 13s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd]
azure_rm_subnet.main: Creating...
azure_rm_subnet.main: Still creating... [00m10s elapsed]
azure_rm_subnet.main: Creation complete after 10s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd/subnets/snet-ad-cyberjkd]
azure_rm_network_interface.main: Creating...
azure_rm_network_interface.main: Creation complete after 6s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd]
azure_rm_network_interface_security_group_association.main: Creating...
azure_rm_network_interface_security_group_association.main: Creation complete after 6s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkSecurityGroups/nsg-ad-cyberjkd]

Error: creating Windows Virtual Machine (Subscription: "62d96334-bbd8-4e04-b5c9-945c8b36f379")
Resource Group Name: "rg-ad-cyberjkd"
Virtual Machine Name: "vm-ad-cyberjkd": performing CreateOrUpdate: unexpected status 409 (409 Conflict) with error: SkuNotAvailable: The requested VM size for resource 'following SKUs have failed for Capacity Restrictions: Standard_DS3_V3' is currently not available in location 'eastus'. Please try another size or deploy to a different location or different zone. See https://aka.ms/azuresknotavailable for details.

with azure_rm_windows_virtual_machine.main,
on main.tf line 83, in resource "azure_rm_windows_virtual_machine" "main":
83: resource "azure_rm_windows_virtual_machine" "main" {

```

Real failure #1 - SkuNotAvailable on the spec'd VM size

■ 10 Check real quota instead of guessing sizes

Run `az vm list-usage --location eastus --output table` to see your subscription's actual quota per VM family, rather than trying sizes one at a time. The next size tried can fail on an entirely different problem - zero quota, not just zero stock.

Expect: A table listing every VM family with *CurrentValue* and *Limit* columns; the failing family shows *Limit: 0*.

```

}
+ virtual_machine_id = (known after apply)
+ vm_agent_platform_updates_enabled = false
+ additional_unattend_content {
  content = (sensitive value)
  setting = "AutoLogon"
}
+ os_disk {
  caching = "ReadWrite"
  disk_size_gb = 127
  name = (known after apply)
  storage_account_type = "Premium_LRS"
  write_accelerator_enabled = false
}
+ source_image_reference {
  offer = "WindowsServer"
  publisher = "MicrosoftWindowsServer"
  sku = "2022-Datacenter"
  version = "latest"
}
+ termination_notification (known after apply)
}
}

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

azurerm_windows_virtual_machine.main: Creating...
Error: creating Windows Virtual Machine (Subscription: "62096334-bbd8-4e04-b5c9-945c8b36f379")
Resource Group Name: "rg-ad-cyberjkd"
Virtual Machine Name: "vm-ad-cyberjkd": performing CreateOrUpdate: unexpected status 409 (409 Conflict) with error: OperationNotAllowed: Operation could not be completed as it results in exceeding approved standardD5v5family Cores quota. Additional details - Deployment Model: Resource Manager, Location: eastus, Current Limit: 0, Current Usage: 0, Additional Required: 2, (Minimum) New Limit Required: 2. Setup Alerts when Quota reaches threshold. Learn more at https://aka.ms/quotaalerting. Submit a request for Quota Increase at https://aka.ms/portal/CR/blade/Microsoft_Azure_Capacity/UsageAndQuota.ReactView/Parameters?SubscriptionId=62096334-bbd8-4e04-b5c9-945c8b36f379&Location=eastus2&QuotaRequest=2&Unit=2&Count=2&Name=2&Value=2&StandardD5v5family&2570X70X70X70 by specifying parameters listed in the 'Details' section for deployment to succeed. Please read more about quota limits at https://docs.microsoft.com/en-us/azure/azure-supportability/per-virtual-machine-quota-requests
with azurerm_windows_virtual_machine.main,
on main.tf line 83, in resource "azurerm_windows_virtual_machine" "main":
83: resource "azurerm_windows_virtual_machine" "main" {

```

Real failure #2 - zero quota on the next VM family tried

■ 11 Switch region if needed

If quota is tight across the board in one region, changing location in terraform.tfvars to a less contended region (this lab used eastus2) is often faster than hunting for one specific working size.

Expect: terraform apply now plans to destroy and recreate the region-tied resources.

■ 12 Recover from state drift

If a mid-apply region change leaves Terraform referencing a resource that was just destroyed (an InvalidResourceReference / 400 error), don't try to patch around it - run terraform destroy to clear the drifted state fully, then a single clean terraform apply from scratch.

Expect: terraform destroy reports "Destroy complete!", then a fresh apply plans a clean 9 resources to add.

```
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes
azure_network_interface_security_group_association.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkSecurityGroups/nsg-ad-cyberjkd]
azure_network_interface_security_group_association.main: Destruction complete after 5s
azure_network_interface.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd]
azure_network_interface.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd, 00m10s elapsed]
azure_network_security_group.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkSecurityGroups/nsg-ad-cyberjkd, 00m10s elapsed]
azure_network_security_group.main: Destruction complete after 13s
azure_network_interface.main: Destruction complete after 13s
azure_public_ip.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/publicIPAddresses/pip-ad-cyberjkd]
azure_virtual_network.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd]
azure_public_ip.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/publicIPAddresses/pip-ad-cyberjkd, 00m10s elapsed]
azure_virtual_network.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd, 00m10s elapsed]
azure_public_ip.main: Destruction complete after 12s
azure_virtual_network.main: Destruction complete after 12s
azure_resource_group.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd]
azure_resource_group.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd, 00m10s elapsed]
azure_resource_group.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd, 00m20s elapsed]
azure_resource_group.main: Destruction complete after 20s
azure_resource_group.main: Creating...
azure_resource_group.main: Still creating... [00m10s elapsed]
azure_resource_group.main: Creation complete after 10s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd]
azure_network_security_group.main: Creating...
azure_public_ip.main: Creating...
azure_virtual_network.main: Creating...
azure_network_security_group.main: Still creating... [00m10s elapsed]
azure_public_ip.main: Still creating... [00m10s elapsed]
azure_virtual_network.main: Still creating... [00m10s elapsed]
azure_public_ip.main: Creation complete after 10s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/publicIPAddresses/pip-ad-cyberjkd]
azure_network_security_group.main: Creation complete after 10s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkSecurityGroups/nsg-ad-cyberjkd]
azure_virtual_network.main: Creation complete after 14s [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd]
azure_network_interface.main: Creating...

Error: creating Network Interface (Subscription: "62d96334-bbd8-4e04-b5c9-945c8b36f379"
Resource Group Name: "rg-ad-cyberjkd"
Network Interface Name: "nic-ad-cyberjkd"): performing CreateOrUpdate: unexpected status 400 (400 Bad Request) with error: InvalidResourceReference: Resource /subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd/subnets/subnet-ad-cyberjkd referenced by resource /subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd was not found. Please make sure that the referenced resource exists, and that both resources are in the same region.

with azure_network_interface.main,
on main.tf line 63, in resource "azure_network_interface" "main":
63: resource "azure_network_interface" "main" {
PS C:\Users\dalia\repos\az-ad-vms
```

Real failure #3 - state drift after the region switch

■ 13 Match VM size to image generation

A newer VM size (like a v7-series) may only boot Gen2 images, while the lab's default image reference is Gen1. If apply fails with a Hypervisor Generation mismatch, switch the source_image_reference sku to its -g2 suffixed version.

Expect: The VM creation step proceeds past the point where it previously failed.

```

+ size = "Standard_D2s_v7"
+ tags = {
+   "project" = "ad-lab"
+ }
+ virtual_machine_id = (known after apply)
+ vm_agent_platform_updates_enabled = false
+ additional_unattend_content {
+   content = (sensitive value)
+   setting = "Autologon"
+ }
+ os_disk {
+   caching = "ReadWrite"
+   disk_size_gb = 127
+   name = (known after apply)
+   storage_account_type = "Premium_LRS"
+   write_accelerator_enabled = false
+ }
+ source_image_reference {
+   offer = "WindowsServer"
+   publisher = "MicrosoftWindowsServer"
+   sku = "2022-Datacenter"
+   version = "latest"
+ }
+ termination_notification (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

azurerm_windows_virtual_machine.main: Creating...

Error: creating Windows Virtual Machine (Subscription: "62096334-bbd8-4e04-b5c9-945c8b36f379"
Resource Group Name: "rg-ad-cyberjkd")
Virtual Machine Name: "vm-ad-cyberjkd": performing CreateOrUpdate: unexpected status 400 (400 Bad Request) with error: BadRequest: The selected VM size 'Standard_D2s_v7' cannot boot Hypervisor Generation '1'.
If this was a Create operation please check that the Hypervisor Generation of the Image matches the Hypervisor Generation of the selected VM Size. If this was an Update operation please select a Hypervisor Gene
ration '1' VM Size. For more information, see https://aka.ms/azuregen2vm

with azurerm_windows_virtual_machine.main,
on main.tf line 83, in resource "azurerm_windows_virtual_machine" "main":
83: resource "azurerm_windows_virtual_machine" "main" {

```

Real failure #4 - Hypervisor Generation mismatch

■ 14 Let the extension run uninterrupted

Once the VM itself is created, the Custom Script Extension installs AD DS and triggers a reboot - this step alone can take 5+ minutes. Don't interrupt it, even if it looks stuck.

Expect: "Apply complete! Resources: X added, 0 changed, 0 destroyed." with `public_ip`, `domain_name`, and `admin_username` in the Outputs.

```

azurerw_virtual_machine.main: Still creating... [02m40s elapsed]
azurerw_virtual_machine.main: Creation complete after 2m49s [id:/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Compute/virtualMachines/vm-ad-cyberjkd]
azurerw_virtual_machine.extension.ad_setup: Creating...
azurerw_virtual_machine.extension.ad_setup: Still creating... [00m10s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [00m20s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [00m30s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [00m40s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [00m50s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [01m00s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [01m10s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [01m20s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [01m30s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [01m40s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [01m50s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [02m00s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [02m10s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [02m20s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [02m30s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [02m40s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [02m50s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [03m00s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [03m10s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [03m20s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [03m30s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [03m40s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [03m50s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [04m00s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [04m10s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [04m20s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [04m30s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [04m40s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [04m50s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [05m00s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [05m10s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [05m20s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [05m30s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [05m40s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [05m50s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [06m00s elapsed]
azurerw_virtual_machine.extension.ad_setup: Still creating... [06m10s elapsed]
azurerw_virtual_machine.extension.ad_setup: Creation complete after 6m15s [id:/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Compute/virtualMachines/vm-ad-cyberjkd/extensions/install-ad-ds]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
admin_username - "admin"
domain_name - "corp.cyberjkd.com"
public_ip - "20.110.134.167"
PS C:\Users\dalla\repos\az-ad-vms>

```

The clean apply - outputs confirm the domain is live

Part 5 - Connect and Verify

■ 15 Wait before RDPing in

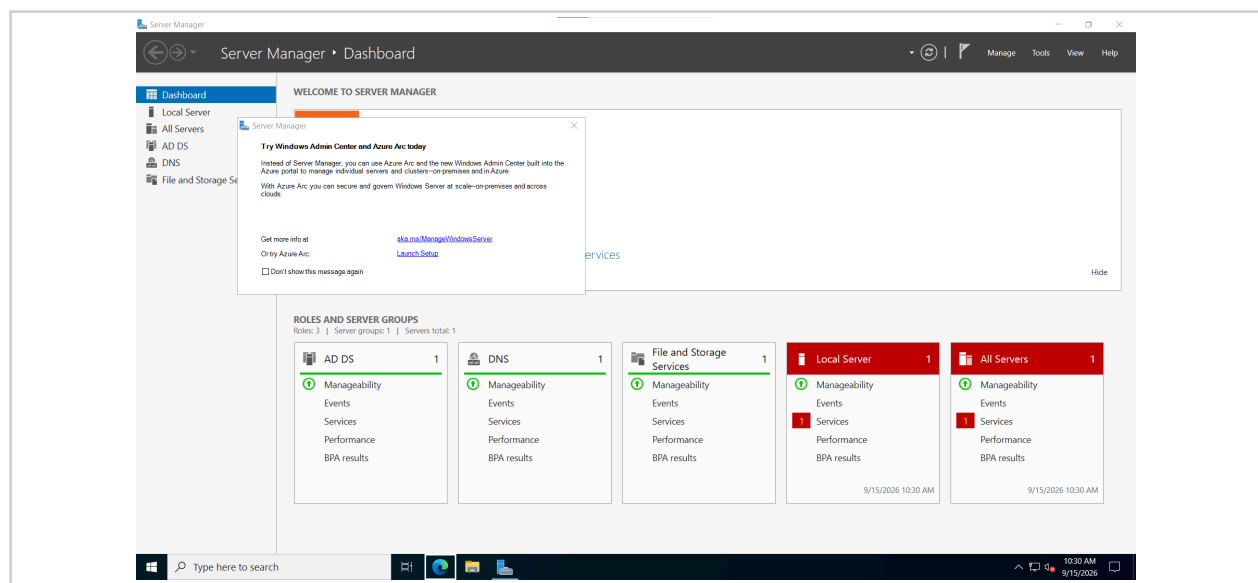
The VM reboots automatically after AD DS installs. Connecting inside the first few minutes usually shows a black screen.

Expect: At least 5 - 10 minutes have passed since apply completed.

■ 16 Connect with the domain credential

Open Remote Desktop Connection, use the public IP from the outputs, and sign in as CORP\adadmin (swap CORP for your own NetBIOS name) - not the local-account syntax.

Expect: Server Manager opens showing AD DS and DNS roles both installed.



Server Manager confirms AD DS and DNS roles are live

■ 17 Run the four verification commands

In an elevated PowerShell on the VM: Get-Service NTDS | Select-Object Name, Status - Get-ADDomain - Get-ADDomainController -Filter * - Resolve-DnsName <yourdomain>

Expect: NTDS shows Running; Get-ADDomain returns a full domain object; the DC is Enabled and a Global Catalog; DNS resolves to the DC's own private IP.

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\adadmin> Get-Service NTDS | Select-Object Name, Status
Name      Status
-----
NTDS      Running

PS C:\Users\adadmin> Get-ADDomain

AllowedDNSSuffixes : {}
ChildDomains       : {}
ComputersContainer : CN=Computers,DC=corp,DC=cyberjkd,DC=com
DeletedObjectsContainer : CN=Deleted Objects,DC=corp,DC=cyberjkd,DC=com
DistinguishedName  : DC=corp,DC=cyberjkd,DC=com
DNSRoot            : corp.cyberjkd.com
DomainControllersContainer : OU=Domain Controllers,DC=corp,DC=cyberjkd,DC=com
DomainMode         : Windows2012Domain
DomainSID          : S-1-5-21-3600838706-3545485195-3249930359
ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=corp,DC=cyberjkd,DC=com
Forest             : corp.cyberjkd.com
InfrastructureMaster : ad-cyberjkd.corp.cyberjkd.com
LastLogonReplicationInterval :
LinkedGroupPolicyObjects : {CN=1182F340-010D-11D2-945F-00C04FB984F9}, CN=Policies, CN=System, DC=corp, DC=cyberjkd, DC=com
LostAndFoundContainer : CN=LostAndFound,DC=corp,DC=cyberjkd,DC=com
ManagedBy         : corp
NetBIOSName        : CYBERJKD
ObjectClass         : domainDNS
ObjectGUID         : 3a558a2-1ed5-47ba-af9d-f870c7fb767
ParentDomain        :
Replicator          : ad-cyberjkd.corp.cyberjkd.com
PublicKeyRequiredPasswordRolling : True
QuotasContainer    : CN=NTDS Quotas,DC=corp,DC=cyberjkd,DC=com
ReadOnlyReplicaDirectoryServers : {}
ReplicaDirectoryServers : {ad-cyberjkd.corp.cyberjkd.com}
RIDMaster          : ad-cyberjkd.corp.cyberjkd.com
SubordinateReferences : {DC=ForestDnsZones,DC=corp,DC=cyberjkd,DC=com, DC=DomainDnsZones,DC=corp,DC=cyberjkd,DC=com, CN=Configuration,DC=corp,DC=cyberjkd,DC=com}
SystemsContainer   : CN=system,DC=corp,DC=cyberjkd,DC=com
UsersContainer      : CN=Users,DC=corp,DC=cyberjkd,DC=com

PS C:\Users\adadmin>

```

NTDS running + full Get-ADDomain output

```
PS C:\Users\adadmin> Get-ADDomainController -Filter *
```

ComputerObjectDN : CN=ad-cyberjkd,CN=Domain Controllers,DC=corp,DC=cyberjkd,DC=com
DefaultPartition : DC=corp,DC=cyberjkd,DC=com
Domain : corp.cyberjkd.com
Enabled : True
Forest : corp.cyberjkd.com
HostName : ad-cyberjkd.corp.cyberjkd.com
InvocationId : 1feab22b-e9fe-4f35-9ddd-7b77245e6778
IPv4Address : 10.0.1.4
IPv6Address :
IsGlobalCatalog : True
IsReadOnly : False
LdapPort : 389
Name : ad-cyberjkd
NTDSSettingsObjectDN : CN=NTDS Settings,CN=ad-cyberjkd,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC=corp,DC=cyberjkd,DC=com
OperatingSystem : Windows Server 2022 Datacenter
OperatingSystemHotfix :
OperatingSystemServicePack :
OperatingSystemVersion : 10.0 (20348)
OperationMasterRoles : {SchemaMaster, DomainNamingMaster, RIDEmulator, RIDMaster...}
Partitions : {DC=ForestDnsZones,DC=corp,DC=cyberjkd,DC=com, DC=DomainDnsZones,DC=corp,DC=cyberjkd,DC=com, CN=Schema,CN=Configuration,DC=corp,DC=cyberjkd,DC=com, CN=Configuration,DC=corp,DC=cyberjkd,DC=com...}
ServerObjectDN : CN=ad-cyberjkd,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC=corp,DC=cyberjkd,DC=com
ServerObjectGuid : 1373ebbd-db41-4451-b91c-7f140a00bdcc
Site : Default-First-Site-Name
SslPort : 636

```
PS C:\Users\adadmin> _
```

Get-ADDomainController - DC enabled, holding all master roles

```
PS C:\Users\adadmin> Resolve-DnsName corp.cyberjkd.com
```

Name	Type	TTL	Section	IPAddress
corp.cyberjkd.com	A	600	Answer	10.0.1.4

```
PS C:\Users\adadmin> _
```

DNS resolves the domain to the DC's own IP

Part 6 - Prove It Actually Works

■ 18 Create an OU, a user, and a group

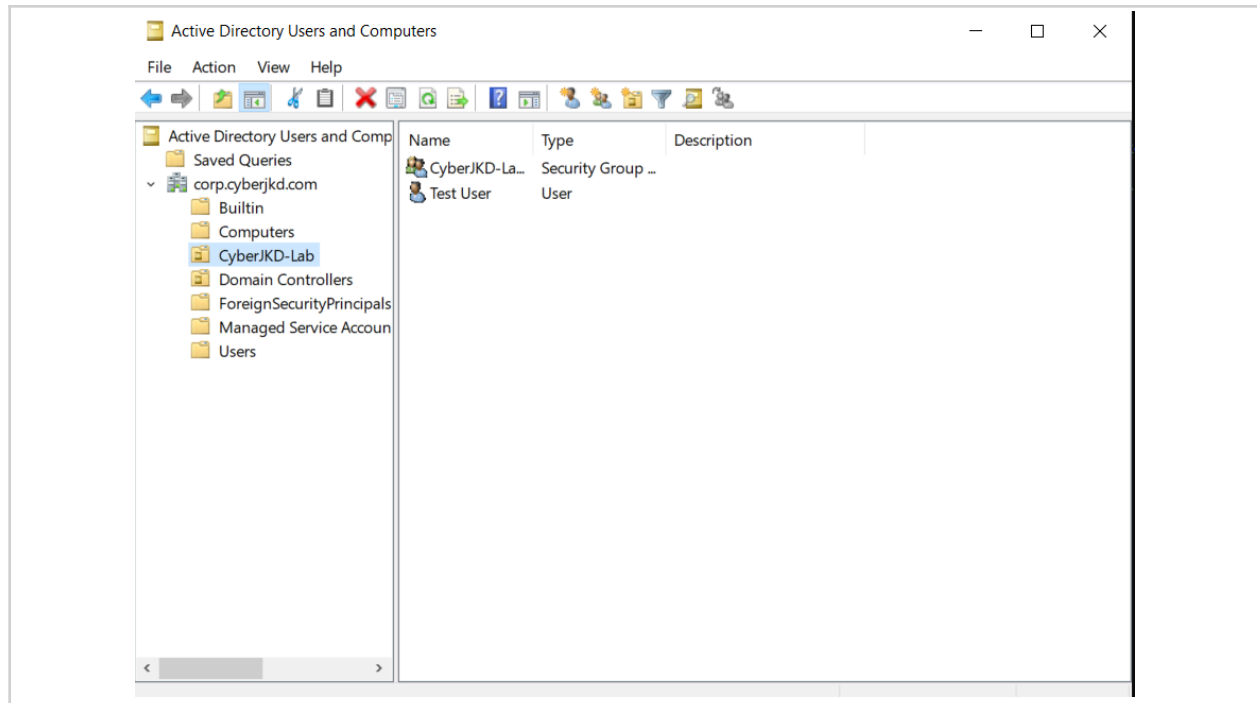
New-ADOrganizationalUnit, New-ADUser, New-ADGroup, then Add-ADGroupMember to place the user inside the group. This is optional relative to the base lab, but it's the difference between "the service is running" and "I can actually manage identity here."

Expect: No errors from any of the four commands.

■ 19 Confirm visually in dsa.msc

Open Active Directory Users and Computers, expand your domain, click your new OU.

Expect: The OU shows both the new user and the new group inside it.



The proof shot - OU containing the user and the group

Part 7 - Teardown

■ 20 Destroy everything

terraform destroy, type yes. This removes every resource this lab created in one operation.

Expect: "Destroy complete! Resources: 9 destroyed."

```

azurerm_windows_virtual_machine.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Compute/virtualMachines/vm-ad-cyberjkd]
azurerm_windows_virtual_machine.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Compute/virtualMachines/vm-ad-cyberjkd, 00m10s elapsed]
azurerm_windows_virtual_machine.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Compute/virtualMachines/vm-ad-cyberjkd, 00m20s elapsed]
azurerm_windows_virtual_machine.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Compute/virtualMachines/vm-ad-cyberjkd, 00m30s elapsed]
azurerm_windows_virtual_machine.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Compute/virtualMachines/vm-ad-cyberjkd, 00m40s elapsed]
azurerm_windows_virtual_machine.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Compute/virtualMachines/vm-ad-cyberjkd, 00m50s elapsed]
azurerm_network_interface.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd]
azurerm_network_interface.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd, 00m10s elapsed]
azurerm_network_interface.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/networkInterfaces/nic-ad-cyberjkd, 00m20s elapsed]
azurerm_network_interface.main: Destruction complete after 22s
azurerm_subnet.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd/subnets/snet-ad]
azurerm_public_ip.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/publicIPAddresses/pip-ad-cyberjkd]
azurerm_subnet.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd/subnets/snet-ad, 00m10s elapsed]
azurerm_public_ip.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/publicIPAddresses/pip-ad-cyberjkd, 00m10s elapsed]
azurerm_public_ip.main: Destruction complete after 18s
azurerm_subnet.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd/subnets/snet-ad, 00m20s elapsed]
azurerm_subnet.main: Destruction complete after 20s
azurerm_virtual_network.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd]
azurerm_virtual_network.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd/providers/Microsoft.Network/virtualNetworks/vnet-ad-cyberjkd, 00m10s elapsed]
azurerm_virtual_network.main: Destruction complete after 19s
azurerm_resource_group.main: Destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd]
azurerm_resource_group.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd, 00m10s elapsed]
azurerm_resource_group.main: Still destroying... [id=/subscriptions/62d96334-bbd8-4e04-b5c9-945c8b36f379/resourceGroups/rg-ad-cyberjkd, 00m20s elapsed]
azurerm_resource_group.main: Destruction complete after 26s

Destroy complete! Resources: 9 destroyed.
PS C:\Users\dalla\repos\az-ad-vm>

```

All 9 resources destroyed cleanly

■ 21 Confirm zero ongoing cost

az resource list --output table

Expect: Only Azure's own auto-created NetworkWatcher entries remain - nothing tied to this lab, nothing billable.

```

PS C:\Users\dalla\repos\az-ad-vm>
PS C:\Users\dalla\repos\az-ad-vm> az resource list --output table

```

Name	ResourceGroup	Location	Type	Status
NetworkWatcher_eastus	NetworkWatcherRG	eastus	Microsoft.Network/networkWatchers	Succeeded
NetworkWatcher_eastus2	NetworkWatcherRG	eastus2	Microsoft.Network/networkWatchers	Succeeded

```

PS C:\Users\dalla\repos\az-ad-vm>

```

Nothing left but Azure's own NetworkWatcher - zero cost

Closing Note

Every failure documented here was a real one - nothing staged. If your own run hits a different error than the ones listed, that's expected too; Azure capacity and quota vary by subscription and by the minute.

The method that matters more than any specific fix is the one used throughout: read the actual error message, check real data (quota tables, SKU restriction lists) before guessing, and treat a failed apply as information rather than a dead end.

 Full roadmap: dallasamuel.github.io/CyberJKD-Roadmap

 All labs: github.com/DallaSamuel/CyberJKD-Labs

*CyberJKD - Becoming dangerous through fundamentals.  *